Learning Frequency-Domain Fusion for Multimodal Remote Sensing Semantic Segmentation

Guangsheng Chen, Fangyu Sun, Weipeng Jing, Weitao Zou, Donglin Di, Yang Song, and Lei Fan

***Abstract*—Multimodal remote sensing data substantially enhance semantic segmentation accuracy by providing complementary information across sensing modalities. However, fully exploiting and effectively fusing features from different modalities to capture comprehensive semantic representations remains a challenge. Most existing methods restrict interactions to the spatial domain, making their representations vulnerable to heterogeneity arising from distinct imaging mechanisms. To address these issues, we propose a frequency-domain decoupled multimodal fusion network (FDMF-Net). Our approach comprises three key modules: the amplitude spectrum decoupling module (ASD), the modality enhancement module (ME), and the low-frequency-guided feature fusion module (LFGF), dedicated to extracting, enhancing, and fusing modality-invariant and specific representations. The ASD module performs frequency-domain decomposition to separate modality-invariant and modality-specific features, promoting more effective cross-modal complementarity. The ME module introduces a mutual information-based feature enhancement scheme to obtain more robust modality-invariant and modality-specific representations, thereby improving feature discriminability. The LFGF module, based on an attention mechanism, fuses shared and specific representations to generate feature maps with richer semantic information. Extensive evaluations on multiple standard multimodal remote sensing datasets demonstrate that FDMF-Net achieves state-of-the-art (SOTA) accuracy across several benchmarks. The code is available at https://github.com/fy-sun/FDMF-Net**

***Index Terms*—Frequency-domain features, multimodal fusion, remote sensing, semantic segmentation.**

I. Introduction

SEMANTIC segmentation is a fundamental yet challenging task in remote sensing image (RSI) interpretation. Driven by rapid advances in satellite-borne observation technology, RSI interpretation has become indispensable for applications such as natural-resource monitoring [1], urban planning [2], and disaster prevention and mitigation [3]. Because images are captured by diverse sensors, RSIs are inherently multimodal, typically including optical, infrared, and synthetic aperture radar (SAR) data. Each modality offers unique, complementary information, e.g., optical images offer rich spectral details, infrared imagery highlights thermal characteristics, and SAR provides all-weather, day–night imaging capability, jointly enriching the feature space for downstream interpretation tasks [4]. Integrating these heterogeneous modalities promises more comprehensive and reliable scene understanding, but it also raises a central question: *how can we integrate multimodal data effectively and efficiently?*

Early work on multimodal RSI segmentation predominantly employed linear fusion strategies, such as feature concatenation [5], addition [6], or inner [7] products. While such straightforward techniques improve segmentation accuracy, they still fall short of fully exploiting the unique properties of each modality [8]. The rapid advance of deep learning, particularly since the introduction of the Transformer architecture [9], has shifted toward adaptive fusion networks equipped with attention mechanisms [10], gating operations [11], and feature alignment modules [12], [13]. Meanwhile, advances in other RSI tasks also validated the effectiveness of adaptive and context-aware fusion strategies. For example, NAS-Kernel [14] integrated multiscale features through a context path aggregation module, thereby significantly improved object counting performance in complex remote sensing scenarios. These networks can dynamically highlight key information and selectively mine complementary features, yet their effectiveness still hinges on how well cross-modal complementarities are modeled [15]. Most existing designs [16] further confine interaction to the spatial domain and overlook the heterogeneity stemming from disparate imaging mechanisms, limiting their ability to tap into truly complementary information. Specifically, optical imagery represents a passive, incoherent measurement of surface reflectance, which is highly sensitive to illumination and atmospheric effects, whereas SAR imagery provides an active, coherent measurement of microwave backscattering, characterized by non-Gaussian speckle noise and geometry-dependent responses. These distinct acquisition processes result in substantial intrinsic heterogeneity between modalities. Conventional spatial-domain fusion methods primarily capture appearance-level correlations under shared local statistics while lacking explicit modeling of sensor-specific information [17]. At the same time, the trend

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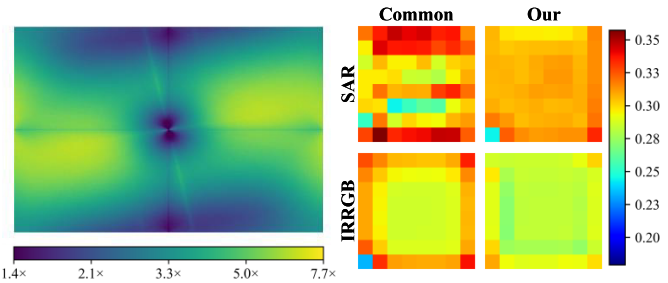
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(a) Relative Amplitude Difference (b) Normalized Mutual Information

Fig. 1. Frequency-domain analysis on WHU-OPT-SAR. (a) Average intermodal relative amplitude spectrum difference across samples. It converts images to luminance, applies DFT, and computes amplitude differences between optical and SAR images. The results showed that low-frequency components were centered and amplitude differences were concentrated in the high-frequency regions. (b) Average NMI between frequency-domain features within each modality. Features are extracted using a four-stage encoder, transformed via DFT, separated into high- and low-frequency components, and fused back with the original spatial features using the IDFT. A higher NMI denotes stronger dependency and thus greater redundancy within the modality. (Left) (Common method) showed high intramodal redundancy, while (right) (our method) significantly reduced such redundancy.

of leveraging vision foundation models in RSI offers a complementary perspective. Gao et al. [18] proposed a change detection framework that combines the segment anything model with limited labeled data, where strong performance was achieved even under few-shot and zero-shot settings. This evidence highlights the importance of introducing priors and efficient adaptation mechanisms under data-scarce conditions, which resonates with our motivation for incorporating frequency-domain priors and information-theoretic constraints to achieve robust multimodal fusion.

Recent studies [19], [20] have verified the effectiveness of frequency-domain learning for single modality RSI segmentation. Unlike conventional spatial-domain approaches, which emphasize local texture and positional cues, frequency-domain features supply both fine details and broad contextual information while suppressing modality-specific noise and sensor artifacts [20]. Several works [21], [22], [23] further extend frequency analysis to multimodal fusion, systematically revealing complementary relationships between the phase and amplitude spectra of SAR and RGB data. Nevertheless, current methods still rely on coarse spectral extraction and simplistic modeling, producing redundant representations that limit the deeper exploitation of cross-modal complementarities and failing to fully harness the synergy between spatial and frequency cues to guide fusion strategies.

We aim to investigate multimodal RSI segmentation from a frequency-domain perspective. To understand where complementary and redundant information resides across different modalities, we begin with an in-depth analysis on the WHU-OPT-SAR dataset, as illustrated in Fig. 1. For each co-registered optical–SAR image pair, we compute the luminance by averaging the red, green, and blue channels and then apply a 2-D discrete Fourier transform (DFT) to obtain the frequency spectra. The amplitude spectra of both images are computed separately, and the relative amplitude difference between them at each frequency is calculated. Low-frequency components (central regions) correspond to slowly varying intensities and global structures, such as large homogeneous areas and coarse layouts, while high-frequency components (periphery/ridges) correspond to edges and fine textures. These differences are averaged across all image pairs to generate the final difference map. The resulting heatmap [Fig. 1(a)] showed that differences were minimal in the low-frequency region (center) but became prominent along high-frequency ridges and corners (periphery). This observation confirms that cross-modal complementarity mainly resides in edge- and texture-rich frequency subbands, consistent with findings in recent work [22]. Furthermore, to assess the redundancy from the coarse-grained spectral extraction used in existing methods, we computed the normalized mutual information (NMI) between high- and low-frequency features within the same modality. Specifically, for either optical or SAR input, we first obtained high- and low-frequency features via an encoder and DFT-based frequency separation with masking (detailed in Section III-B) and then calculated the NMI between the two branches. A higher NMI indicates a larger amount of shared information and stronger correlation. As shown in Fig. 1(b), both modalities exhibited considerable redundancy (NMI $>$ 0.3). In contrast, our proposed method significantly reduced these correlations, producing more compact and decorrelated representations.

Motivated by these insights, we propose a frequency-domain decoupled multimodal fusion network (FDMF-Net) for multimodal RSI semantic segmentation. The central idea is to decouple each modality into mutually noninterfering low- and high-frequency components. The high-frequency branch captures modality-specific details, whereas the low-frequency branch encodes modality-invariant information that guides fusion. Concretely, FDMF-Net comprises three key components: the amplitude spectrum decoupling module (ASD), which adaptively separates amplitude spectrum features into high- and low-frequency representations for each modality. The modality enhancement module (ME) learns a pair of clean and complementary modality-invariant and specific representations for each modality, preventing modality-specific noise from contaminating the invariant branch. Here, the modality-invariant representation captures shared semantic information across modalities, whereas the modality-specific representation characterizes fine details tied to the particular imaging mechanism. Finally, the low-frequency-guided feature fusion module (LFGF) fuses features guided by invariant representations, producing a unified multimodal feature for decoding. Our contributions are as follows.

1) A FDMF-Net has been developed. FDMF-Net enhances modality representations extracted in the frequency domain to fully exploit shared and complementary information across modalities. Modality-invariant cues guide the selective fusion of these enriched features, enabling the network to capture both comprehensive and fine-grained semantic details effectively.
2) The introduction of ASD allows the network to attend to relevant information across frequency bands. ME leverages mutual information to enhance shared and

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complementary representations, providing the network with more intrinsic cross-modal interactions. LFGF employs an attention mechanism to facilitate interactions between modality representations, enhancing feature learning and integrating cues.

3) Extensive experiments were conducted on four public datasets: WHU-OPT-SAR [10], YESeg-OPT-SAR [24], Vaihingen [25], and Potsdam [25], which cover diverse modality combinations, validating the effectiveness of FDMF-Net. For example, on WHU-OPT-SAR, our model achieved 71.66% mean intersection over union (mIoU), surpassing state-of-the-art (SOTA) methods and yielding higher segmentation accuracy on small objects.

II. Related Work

A. Multimodal RSI Semantic Segmentation

Semantic segmentation is a critical preprocessing step in RSI analysis. Leveraging multisource data significantly enhances both accuracy and robustness, as multimodal inputs provide richer spectral, textural, and geometric cues compared to any single modality. A straightforward strategy is a linear combination, wherein features from different modalities are blended directly. For instance, Gupta et al. [26] stacked RGB and depth channels as input to a convolutional neural network (CNN). Zhang et al. [27] exploited a block-based convolutional architecture to jointly process optical and SAR data, while Lin et al. [6] built a unified data cube from preprocessed optical–SAR pairs for whole-image classification. Another popular line of work employs a dual-branch network to extract features separately from each modality before hierarchical fusion. Hughes et al. [28] adopted a pseudo-Siamese CNN to process optical and SAR data independently, followed by feature concatenation. While these approaches are computationally efficient, they typically perform shallow-level fusion and apply uniform weighting to all inputs. However, they struggle to enhance complementary cues while suppressing redundant noise, thereby limiting the discriminative power of the fused features.

Recent studies have leveraged more sophisticated deep learning-based fusion methods. MCANet [10] employs a self-attention (SA) mechanism to model spatial relationships between cross-modal features, enabling rich optical–SAR interactions. Wang et al. [29] developed a channel-attention-driven network that dynamically weights modality-specific information. CHGFNet [30] employs a complementary gating mechanism to adaptively merge multimodal inputs, thereby enhancing segmentation accuracy. Other studies focus on aligning feature distributions across modalities to facilitate more effective fusion. For example, Li et al. [31] projected both modalities into a shared latent space to align their semantic distributions.

Despite these advances, current methods perform fusion mainly in the spatial domain. Given that optical and SAR sensors are based on fundamentally different imaging mechanisms, spatial-level fusion alone cannot fully capture their cross-modal complementarity. To overcome this limitation, we propose FDMF-Net, which introduces a frequency-domain perspective. By jointly integrating frequency and spatial features, our model more effectively exploits cross-modal complementarity, leading to improved segmentation accuracy.

B. Frequency-Domain Learning

Frequency-domain analysis, a cornerstone of traditional signal processing [32], has recently gained renewed attention in deep learning frameworks, where it supports the learning of nonlocal, structure-aware features [33], [34], [35], [36], [37]. Frequency-based methods have been widely applied to various computer vision tasks. For instance, SFINet [38] combines high spatial resolution guidance in the spatial domain with low resolution target images in the frequency domain, achieving competitive performance in two representative multimodal image fusion tasks: pan sharpening and depth super-resolution. Chen et al. [39] proposed an amplitude–phase recombination strategy, wherein the amplitude spectrum of source images is replaced with that of the target domain while preserving the source phase, enabling cross-domain sample synthesis and improved model generalization. In dense prediction tasks, FDBANet [40] introduces frequency-domain attention to denoise sonar imagery and strengthen feature representations, thereby improving segmentation accuracy. FreqFusion [41] further develops adaptive frequency filters, where an adaptive low-pass filter (ALPF) smooths high-frequency noise to unify intraclass features, while an adaptive high-pass filter (AHPF) recovers critical boundary information lost during downsampling. Similarly, Chen et al. [42] showed that low-pass filtering effectively suppresses high-frequency feature noise in images, improving instance segmentation under low-light conditions.

In the context of RSI semantic segmentation, frequency-domain analysis has also demonstrated promising improvements. SFFNet [43] bridges the semantic gap between spatial and frequency features via wavelet transforms, enhancing salient feature selection and enabling the effective combination of representations from different domains. Meanwhile, AFENet [20] adaptively generates amplitude spectrum masks during the decoding stage to retain optimal details and contextual information for land-cover feature representation. Furthermore, FDNet [44] decouples and independently optimizes high- and low-frequency components in the frequency domain, effectively combining CNN-based spatial feature extraction with frequency-domain contextual refinement to achieve superior performance in complex RSI scenarios. Similarly, FGNet [45] incorporates frequency-domain enhancement into the feature extraction stage, strengthening the joint representation of global–local and spatial–frequency cues. Recent work in multimodal semantic segmentation also emphasizes the role of frequency components. Li et al. [21] treated phase information as shared across modalities and incorporated it into intermediate layers to facilitate cross-modal fusion. PAD [22] achieves fusion through phase–amplitude decoupling, revealing that phase components are largely shared across modalities while amplitude captures modality-specific content.

Despite these advances, most prior approaches fuse frequency-domain representations without explicitly addressing their intrinsic redundancy. Moreover, cross-modal

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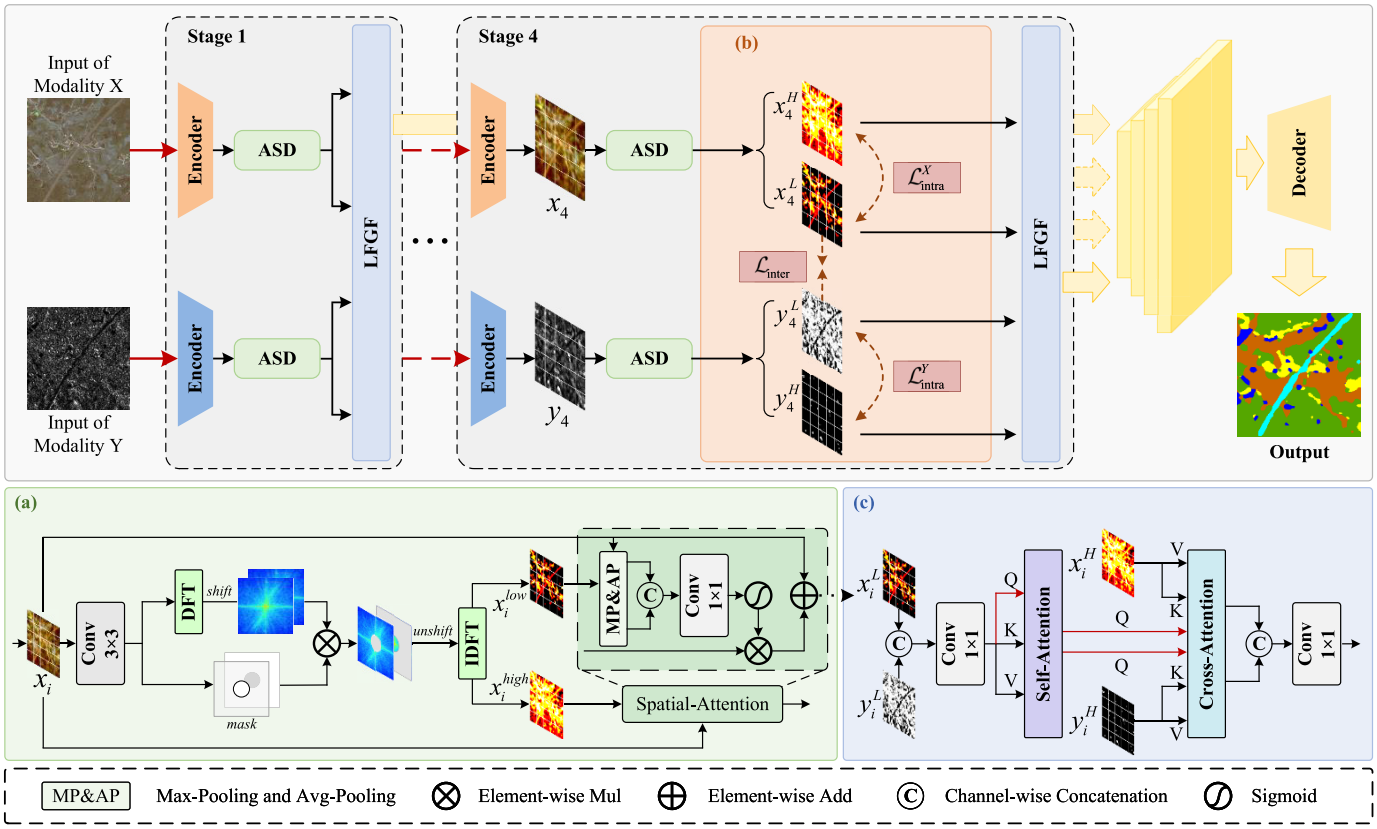


Fig. 2. Overview of the proposed FDMF-Net. It consists of three key modules. (a) ASD. (b) ME. (c) LFGF. For input image pairs of modalities X and Y, denoted (x, y), each stage $i \in \{1, 2, 3, 4\}$ first extracts features (x_i, y_i) via the respective encoder, which is then processed by the ASD module to decouple high- and low-frequency components. At the final stage ($i = 4$), the ME module refines these frequency-domain features to produce modality-invariant and modality-specific representations. Finally, the LFGF module uses the modality-invariant cues to guide the interaction and fusion of complementary information. The fused output from each stage is then fed into a lightweight shared decoder to generate the final segmentation.

relationships among frequency components are underexplored, limiting their fusion efficacy. To this end, we propose FDMF-Net, which begins with learnable frequency separation and masking to extract high- and low-frequency features. Unlike prior works, we explicitly reduce intramodal redundancy via mutual information constraints, yielding complementary and compact modality-specific high-frequency representations. Meanwhile, we selectively enhance low-frequency signals to better capture cross-modal shared information. Finally, these enhanced low-frequency features serve as modality-invariant anchors to guide the interaction and fusion of modality-specific high-frequency representations, thereby enabling more effective multimodal feature integration.

III. Methodology

A. Overview

We introduce a FDMF-Net designed for semantic segmentation in RSI. As shown in Fig. 2, our architecture integrates a dual-branch encoder to extract modality-specific features and a shared decoder to reconstruct semantic predictions. Given input modalities $x, y \in \mathbb{R}^{C\times H\times W}$ for modalities X and Y, respectively (e.g., optical and SAR images), where C, H, and W denote channels, height, and width, respectively, each modality is processed by an independently parameterized encoder to extract multiscale feature representations. The encoder outputs are hierarchical features from four resolution stages $\{1/4, 1/8, 1/16, 1/32\}$ of the original input size, corresponding to stages $i \in \{1, 2, 3, 4\}$. The encoders update features via

$$x_i = E_x^{(i)}(x_{i-1}), \quad y_i = E_y^{(i)}(y_{i-1}) \tag{1}$$

where $E_x^{(i)}$ and $E_y^{(i)}$ denote the ith encoder blocks for modalities X and Y, respectively. For the first stage ($i = 1$), raw inputs are directly used: $x_0 = x$ and $y_0 = y$.

Within each stage, features x_i and y_i are first passed through the ASD, which adaptively decouples them into high-frequency (x_i^H and y_i^H) and low-frequency components (x_i^L and y_i^L) in the frequency domain. These components are then gated and propagated to the next encoder stage. The ME further refines these frequency components via mutual information objectives, where it encourages low-frequency features to align across modalities as shared semantic content while preserving the uniqueness of high-frequency features. Concretely, the enhanced low-frequency features serve as modality-invariant representations, capturing cross-modal stable semantics such as overall layout, region boundaries, and slowly varying structures that are insensitive to illumination or acquisition conditions. In contrast, the high-frequency features act as modality-specific representations, characterizing sensor-dependent details such as fine textures in optical imagery or speckle-related microstructures in SAR imagery.

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This decoupling enhances both cross-modal fusion and robustness to redundant modality-specific noise. The LFGF is employed to fuse cross-modal information. It fuses the modality-invariant (low-frequency) signals while selectively incorporating complementary modality-specific (high-frequency) features, resulting in fused semantic features f_i^{fused}. Finally, following the SegFormer architecture, we adopt a lightweight MLP-based decoder. This decoder upsamples high-level features through interpolation, concatenates them with shallow features from earlier encoder layers, and performs final refinement to produce dense prediction maps at full resolution.

B. Amplitude Spectrum Decoupling Module

To capture low- and high-frequency information in RSIs, we propose the ASD module, which consists of two key components: adaptive spectral separation and spatial frequency-domain enhancement. Here, we illustrate the procedure using modality X, and modality Y follows an identical process.

1) Adaptive Spectral Separation: Given the spatial-domain feature $x_i \in \mathbb{R}^{C_i \times H_i \times W_i}$ generated by the ith encoder stage, we first employ a DFT, denoted as $\mathcal{F}(\cdot)$, to map it to the frequency domain and obtain the complex-valued spectrum $s_i = \mathcal{F}(x_i) \in \mathbb{C}^{C_i \times H_i \times W_i}$. The transformation is defined as

$$s_i(u,v) = \frac{1}{H_i W_i} \sum_{h=0}^{H_i-1} \sum_{w=0}^{W_i-1} x_i(h,w)\, e^{-j2\pi\left(\frac{uh}{H_i}+\frac{vw}{W_i}\right)} \tag{2}$$

where (u, v) denotes the frequency coordinates. Subsequently, an FFT-shift operation $\mathcal{S}(\cdot)$ relocates the low-frequency components to the center of the spectrum, facilitating the construction of a frequency-domain binary mask $\mathbf{M}_i \in \mathbb{R}^{H_i \times W_i}$

$$\mathbf{M}_i^{(u,v)} = \begin{cases} 1, & |u| \le r_h, \quad |v| \le r_w \\ 0, & \text{otherwise} \end{cases} \tag{3}$$

where r_h and r_w are learnable parameters defining the low-frequency range. Using this mask, we decompose the frequency-domain spectrum into low- and high-frequency components

$$s_i^{\text{low}} = \mathcal{S}(s_i) \odot \mathbf{M}_i, \quad s_i^{\text{high}} = \mathcal{S}(s_i) \odot (1 - \mathbf{M}_i) \tag{4}$$

where $\odot$ represents the Hadamard product. We then apply the inverse FFT-shift $\mathcal{S}^{-1}(\cdot)$ and IDFT $\mathcal{F}^{-1}(\cdot)$ to map these components back to the spatial domain. Only the amplitude is retained, yielding the spatial-domain representations

$$x_i^{\text{low}} = \left|\mathcal{F}^{-1}\left(\mathcal{S}^{-1}\left(s_i^{\text{low}}\right)\right)\right|, \quad x_i^{\text{high}} = \left|\mathcal{F}^{-1}\left(\mathcal{S}^{-1}\left(s_i^{\text{high}}\right)\right)\right| \tag{5}$$

where $|\cdot|$ denotes the absolute value operator. Although geometric and positional information is predominantly carried by the phase, modality-specific contrast and fine textures are primarily reflected in the amplitude spectrum. Therefore, after frequency decomposition, we retain only the amplitude component, which serves as a stable real-valued cue for selective enhancement between the spatial and frequency domains. At the same time, the structural information encoded in the phase is not discarded. Instead, by incorporating the original spatial-domain signal x_i and applying spatial attention to re-emphasize edges and boundaries, we effectively restore the geometric structures associated with the phase.

2) Spatial Frequency-Domain Enhancement: The low-frequency feature x_i^{low} encodes global context and structural information, whereas the high-frequency feature x_i^{high} emphasizes local details such as textures and edges. To bridge their semantic gap and boost their representation power, we introduce a spatial attention mechanism that enhances both components individually. For the low-frequency feature x_i^{low}, we compute attention weights as

$$\mathbf{F}_i^L = \sigma\left(\mathbf{Conv}\left(\left[\mathcal{A}(x_i), \mathcal{M}(x_i), \mathcal{A}\left(x_i^{\text{low}}\right), \mathcal{M}\left(x_i^{\text{low}}\right)\right]\right)\right) \tag{6}$$

where $\mathcal{A}(\cdot)$ and $\mathcal{M}(\cdot)$ represent channel-wise average and max pooling, respectively; $[\cdot]$ denotes channel-wise concatenation; $\sigma(\cdot)$ is the sigmoid activation function; and $\mathbf{Conv}(\cdot)$ refers to a 1×1 convolutional layer. The enhanced low-frequency feature is then obtained by $x_i^L = x_i + \mathbf{F}_i^L \times x_i^{\text{low}}$. The high-frequency branch follows the same procedure, substituting x_i^{low} with x_i^{high}, yielding enhanced features x_i^H. The same operations are performed on modality Y, ultimately producing four enhanced feature maps: $x_i^L, x_i^H, y_i^L, y_i^H \in \mathbb{R}^{C_i \times H_i \times W_i}$ for (x_i, y_i). These serve as frequency-aware, decoupled representations for subsequent fusion and decoding stages.

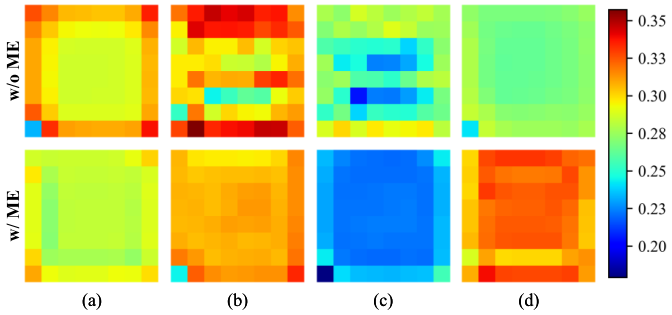


Fig. 3. Average NMI heatmaps for high- and low-frequency representations within and across modalities on the WHU-OPT-SAR dataset. (a) Intramodal NMI between high- and low-frequency representations, with redundancy concentrated at the edges of the feature map and significantly reduced by the ME module. (b) Intramodal NMI for the SAR modality, showing irregular redundancy patterns that the ME module effectively suppresses. (c) Intermodal NMI for high-frequency representations, which should remain low for modality-specific cues and is successfully maintained by ME. (d) Intermodal NMI for low-frequency representations, where ME produces more cohesive modality-invariant features.

C. Modality Enhancement Module

Although the ASD module extracts high- and low-frequency components, significant redundancy persists between them. As shown in Fig. 3(a) and (b), both modalities in the WHU-OPT-SAR dataset exhibit high NMI between their high- and low-frequency branches, indicating substantial intramodal overlap. Such redundancy hampers the learning of truly complementary modality-specific and modality-invariant representations, which in turn weakens downstream segmentation performance.

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Moreover, the low-frequency features of the two modalities exhibit weak correlation, despite the fact that low-frequency cues are expected to encode modality-invariant structural semantics. Simultaneously, the high-frequency branches exhibit strong intermodal similarity, which highlights their capacity to capture modality-specific textures and details. As shown in Fig. 3(c) and (d), the NMI between high-frequency features across modalities remains elevated, indicating persistent redundancy, while consistency among low-frequency features is clearly inadequate. To alleviate these issues, we introduce the ME module that: 1) minimizes the mutual information $\mathrm{MI}(x_i^L, x_i^H)$ between a modality's own low- and high-frequency features and 2) maximizes the mutual information $\mathrm{MI}(x_i^L, y_i^L)$ between the low-frequency features of different modalities. This dual objective drives the network to learn complementary, nonredundant high- and low-frequency cues while reinforcing the low-frequency information shared across modalities, thereby decoupling modality-invariant content (low-frequency features) from modality-specific details (high-frequency features).

1) Intramodal Minimization: Since directly optimizing mutual information is intractable, we instead employ the contrastive log-ratio upper bound (CLUB) and its variational extension (vCLUB) as a differentiable surrogate objective [46]. Given sample pairs $\{(x_n, y_n)\}_{n=1}^N$, vCLUB is defined as

$$\begin{aligned} I_{\mathrm{vCLUB}}(x,y) &:= \mathbb{E}_{p(x,y)}\left[\log q_\theta(y|x)\right] - \mathbb{E}_{p(x)}\mathbb{E}_{p(y)}\left[\log q_\theta(y|x)\right] \\ &= \frac{1}{N^2}\sum_{n=1}^{N}\sum_{m=1}^{N}\left[\log q_\theta(y_n|x_n) - \log q_\theta(y_m|x_n)\right] \end{aligned} \tag{7}$$

where $q_\theta(y|x)$ is a variational distribution parameterized by θ that approximates the true conditional distribution $p(y|x)$. Here, $\log q_\theta(y_n|x_n)$ denotes the log likelihood for the positive pair (x_n, y_n), while $\{\log q_\theta(y_m|x_n)\}_{n\neq m}$ provides the log likelihood of the negative pair (x_n, y_m). To tighten the bound, we maximize the variational log likelihood

$$\mathcal{L}_{q_\theta}(x,y) = \frac{1}{N}\sum_{n=1}^{N}\log q_\theta(y_n|x_n). \tag{8}$$

We define the intramodal loss $\mathcal{L}_{\mathrm{intra}} = \mathcal{L}_{\mathrm{intra}}^X + \mathcal{L}_{\mathrm{intra}}^Y$ as a minimization objective, explicitly reducing the mutual information between the high- and low-frequency features within each modality, where

$$\begin{cases} \mathcal{L}_{\mathrm{intra}}^X = I_{\mathrm{vCLUB}}\left(x_i^L, x_i^H\right) - \mathcal{L}_{q_\theta}\left(x_i^L, x_i^H\right) \\ \mathcal{L}_{\mathrm{intra}}^Y = I_{\mathrm{vCLUB}}\left(y_i^L, y_i^H\right) - \mathcal{L}_{q_\theta}\left(y_i^L, y_i^H\right) \end{cases}, \quad i = 4. \tag{9}$$

Here, $i = 4$ indicates that the ME module computes mutual information loss on only the output of the final encoder. Performing mutual information optimization at the final encoder stage allows for more effective decoupling of modal-shared and modality-specific information. Moreover, it leverages the specific and discriminative power of deep features while avoiding constraining shared information in shallow features.

In practice, the variational distribution $q_\theta(y|x)$ is typically parameterized by an MLP that learns both the conditional mean and variance. For greater stability and efficiency, we adopt a simplified variant, often referred to as vCLUB-MEAN, which predicts only the conditional mean while fixing the variance to 1. Specifically, we first apply global average pooling to convert the spatial features into 2-D vectors, followed by an MLP to predict the conditional mean. Negative pairs are constructed by shuffling samples within the mini-batch. During training, we jointly minimize the vCLUB upper bound and maximize the variational log likelihood, consistent with (9).

2) Intermodal Maximization: To ensure that the low-frequency features across different modalities encode consistent semantic information, we aim to maximize the mutual information between them. To this end, we employ the trainable lower bound introduced by MINE [47] to approximately maximize mutual information. Given paired samples $\{(x_n, y_n)\}_{n=1}^N$, the MINE objective is defined as

$$\begin{aligned} I_{\mathrm{MINE}}(x,y) &:= \mathbb{E}_{p(x,y)}\left[T_\Omega\right] - \log\left(\mathbb{E}_{p(x)}\mathbb{E}_{p(y)}\left[e^{T_\Omega}\right]\right) \\ &= \frac{1}{N}\sum_{n=1}^{N} T_\Omega(x_n, y_n) \\ &\quad - \log\left(\frac{1}{N^2}\sum_{n=1}^{N}\sum_{m=1}^{N}\exp\left(T_\Omega(x_n, y_m)\right)\right) \end{aligned} \tag{10}$$

where T_Ω denotes a neural network with trainable parameters Ω that estimates the mutual information score between input pairs. The first term reflects the average score of positive pairs (x_n, y_n), while the second term computes the log-partition function over all negative (nonmatching) pairs. To encourage the model to learn modality-invariant features in the low-frequency branches, we define the intermodal loss $\mathcal{L}_{\mathrm{inter}}$ as

$$\mathcal{L}_{\mathrm{inter}} = -I_{\mathrm{MINE}}\left(x_i^L, y_i^L\right), \quad i = 4 \tag{11}$$

where the negative sign transforms the maximization into a minimization objective. As with the intramodal loss, this is applied only at the final encoder stage ($i = 4$), where semantic features are most informative and aligned across modalities. The implementation of MINE follows a similar design. We first apply global average pooling to obtain compact 2-D vectors and construct negative pairs by shuffling batch samples across modalities. A two-layer MLP is employed as the critic network $T_\Omega(\cdot)$ to compute the mutual information score. This score is maximized to reinforce the semantic consistency of low-frequency features across modalities.

The final objective of the ME module combines both $\mathcal{L}_{\mathrm{intra}}$ and $\mathcal{L}_{\mathrm{inter}}$, which is computed as

$$\mathcal{L}_{\mathrm{ME}} = \alpha\mathcal{L}_{\mathrm{intra}} + \beta\mathcal{L}_{\mathrm{inter}} \tag{12}$$

where hyperparameters α and β control the relative contributions of the intramodal and intermodal mutual information terms. This dual constraint encourages the network to learn more compact, complementary high-frequency features and semantically consistent low-frequency features.

D. Low-Frequency-Guided Feature Fusion Module

Following the ME module, we obtain modality-invariant low-frequency representations $\{x_i^L, y_i^L\}$ and modality-specific high-frequency representations $\{x_i^H, y_i^H\}$. To effectively integrate this information, we introduce the LFGF module, which

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leverages shared low-frequency cues to guide the integration of complementary high-frequency details from the two modalities. Specifically, we first concatenate the low-frequency features to obtain a unified modality-invariant representation f_i^L. Next, we employ a Transformer-based fusion block, which includes both SA and cross-attention (CA) operations. The SA blocks on the modality-invariant representation distill global context from the unified low-frequency embedding (LF), while the CA blocks capture intermodality interactions between the low- and high-frequency features. Let $z \in \mathbb{R}^{L\times d}$ denote the matrix obtained by flattening f_i^L along its spatial dimensions. The SA block on input z is defined as

$$\mathrm{SA}(z) = Q^z + \mathbf{MLP}\left(Q^z + \mathrm{softmax}\left(\frac{Q^z K^{z\top}}{\sqrt{d_k}}\right)V^z\right) \quad (13)$$

where $Q^z = z\mathbf{W}^Q \in \mathbb{R}^{L\times d_k}$, $K^z = z\mathbf{W}^K \in \mathbb{R}^{L\times d_k}$, and $V^z = z\mathbf{W}^V \in \mathbb{R}^{L\times d_v}$; $\mathbf{W}^Q, \mathbf{W}^K \in \mathbb{R}^{d\times d_k}$ and $\mathbf{W}^V \in \mathbb{R}^{d\times d_v}$ are learnable projection matrices. This operation refines the modality-invariant representation via global context modeling. The CA block uses the LF $l = \mathrm{SA}(z)$ as the query and the high-frequency embedding $h \in \{h_x, h_y\}$ as the key/value, where h_x and h_y are obtained by flattening the high-frequency maps x_i^H and y_i^H, respectively. This enables modality-invariant cues to selectively retrieve complementary high-frequency details

$$\mathrm{CA}(l,h) = Q^l + \mathbf{MLP}\left(Q^l + \mathrm{softmax}\left(\frac{Q^l K^{h\top}}{\sqrt{d_k}}\right)V^h\right) \quad (14)$$

where $Q^l = l\mathbf{W}^Q$, $K^h = h\mathbf{W}^K$, and $V^h = h\mathbf{W}^V$. The resulting $\mathrm{CA}(l, h_x)$ and $\mathrm{CA}(l, h_y)$ are concatenated along the channel dimension and then processed by a convolutional layer, yielding the fused representation f_i^{fused}. By leveraging modality-invariant low-frequency features as anchors, the LFGF module selectively integrates high-frequency details from both modalities. This mechanism enhances semantic consistency while preserving complementary discriminative cues, ultimately delivering richer feature maps to the shared decoder.

E. Loss Function

We employ a composite loss function $\mathcal{L}_{\mathrm{total}}$ that combines cross-entropy loss $\mathcal{L}_{\mathrm{ce}}$, Dice loss $\mathcal{L}_{\mathrm{dice}}$, and the ME loss $\mathcal{L}_{\mathrm{ME}}$, which is computed as

$$\mathcal{L}_{\mathrm{total}} = \mathcal{L}_{\mathrm{ce}} + \lambda_1 \mathcal{L}_{\mathrm{ME}} + \lambda_2 \mathcal{L}_{\mathrm{dice}} \quad (15)$$

where λ_1 and λ_2 are hyperparameters. The cross-entropy loss is defined as

$$\mathcal{L}_{\mathrm{ce}} = -\frac{1}{N}\sum_{n=1}^{N}\sum_{k=1}^{K} g_n^k \log \hat{g}_n^k \quad (16)$$

where N is the number of pixels, K is the number of classes, g_n^k is the ground truth one-hot label, and $\hat{g}_n^k$ is the predicted probability for class k at pixel n. To mitigate class imbalance, we incorporate the Dice loss

$$\mathcal{L}_{\mathrm{dice}} = 1 - \frac{2\sum_{n=1}^{N}\sum_{k=1}^{K}\hat{g}_n^k g_n^k}{\sum_{n=1}^{N}\sum_{k=1}^{K}\left(\hat{g}_n^k + g_n^k\right)}. \quad (17)$$

TABLE I
STATISTICS OF THE FOUR MULTIMODAL BENCHMARK DATASETS

Dataset	Modality Type	Training Samples	Test Samples	# of Categories
WHU-OPT-SAR	IRRGB-SAR	8000	2000	7
YESeg-OPT-SAR	RGB-SAR	1784	447	7
ISPRS Vaihingen	IRRG-DSM	10000	266	5
ISPRS Potsdam	IRRGB-DSM	10000	3174	5

By jointly combining three terms, our model effectively balances pixel-level classification, region-level segmentation quality, and the learning of disentangled and informative multimodal representations.

IV. Experimental Results

A. Datasets

1) WHU-OPT-SAR [10]: This dataset contains 100 optical images of 5556×3704 pixels and corresponding single-band SAR images, all acquired over Hubei Province in China. The optical images consist of four bands: red, green, blue, and near infrared (NIR). Each image is annotated with seven primary land use categories: farmland, city, village, water, forest, road, and others. For preprocessing, all images were cropped into nonoverlapping patches of 256×256 pixels, resulting in 10 000 total patches. We adopted an 80%/20% train–test split.

2) YESeg-OPT-SAR [24]: This dataset extends multimodal data to coastal environments. It consists of 2231 pairs of high-resolution optical and SAR tiles, each with 256×256 pixels. Every pixel is exhaustively labeled as one of seven semantic categories: bare ground, low vegetation, trees, houses, water, roads, and others. The dataset was divided into 80% for training and 20% for testing.

3) ISPRS Vaihingen [25]: This dataset contains 16 orthophotos, each averaging 2500×2000 pixels, from the Vaihingen region in Germany. These orthophotos include three bands: NIR, red, and green, along with normalized digital surface models (nDSMs). Ground truth annotations cover five foreground classes: impervious surfaces, buildings, low vegetation, trees, and cars. The orthophotos were split into a training set of 12 images and a test set of four images. Instead of cropping, we applied a 256×256 sliding window during training and testing.

4) ISPRS Potsdam [25]: This dataset contains 24 orthophotos of size 6000×6000 pixels from Potsdam, Germany. These images feature four bands: red, green, blue, and NIR, alongside nDSMs. The dataset includes annotations for the same foreground classes as the Vaihingen dataset. The orthophotos were divided into 18 images and 6 images for training and testing, respectively. Similar to Vaihingen, a 256×256 sliding window was used to generate training and test patches.

B. Implementation Details

All experiments were conducted using PyTorch on a single NVIDIA A100 GPU with 125 GB of RAM. To accelerate convergence, we initialized our model's encoder with SegFormer (MiT-B4) [48]. We optimized our models using the AdamW optimizer with a learning rate of 1×10^{-4} and a cosine

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TABLE II

QUANTITATIVE RESULTS (%) ON WHU-OPT-SAR AND YESEG-OPT-SAR, WITH THE MEAN PERFORMANCE ACROSS BOTH OPTICAL AND SAR MODAL DATASETS REPORTED AS "MEAN." THE BEST RESULTS ARE IN BOLD. THE SECOND-BEST RESULTS ARE UNDERLINED. * INDICATES VARIANCE <0.1 OVER THREE RUNS, AND † INDICATES VARIANCE <0.5

	Method	Backbone	WHU-OPT-SAR				YESeg-OPT-SAR				Mean			
			OA	Kappa	mF1	mIoU	OA	Kappa	mF1	mIoU	OA	Kappa	mF1	mIoU
UniModal	SegFormer [48]	SegFormer-B4	88.86	83.93	70.67	60.00	84.69	72.43	72.02	60.54	86.78	78.18	71.35	60.27
	TransUNet [49]	R50-ViT-B	89.08	84.36	71.53	60.37	84.28	72.03	75.99	63.04	86.68	78.20	73.76	61.71
	UNetFormer [50]	ResNet-18	89.41	84.87	70.82	60.15	85.16	73.87	70.68	59.16	87.29	79.37	70.75	59.66
	RS^{3}Mamba [51]	R18-Mamba-T	89.34	84.75	71.38	60.46	85.34	74.01	69.10	58.75	87.34	79.38	70.24	59.61
	PyramidMamba [52]	ResNet-18	88.64	83.81	70.87	59.45	85.36	74.17	74.56	61.89	87.00	78.99	72.72	60.67
	SFFNet [43]	ConvNext-T	89.26	84.62	71.30	60.35	83.03	70.09	73.75	60.67	86.15	77.36	72.53	60.51
	AFENet [20]	ResNet-18	88.97	84.20	70.46	59.55	85.19	73.50	77.51	64.66	87.08	78.85	73.99	62.11
MultiModal	MCANet [10]	ResNet-101	89.97	85.76	71.11	60.63	84.65	72.60	74.57	61.41	87.31	79.18	72.84	61.02
	CMFNet [53]	VGG-16	91.49	87.85	75.11	64.85	84.29	72.32	66.97	54.81	87.89	80.09	71.04	59.83
	CMX [54]	SegFormer-B2	92.89	89.85	79.18	69.22	**88.22**	**79.31**	81.85	70.50	**90.56**	**84.58**	80.52	69.86
	MultiSenseSeg [29]	ResNet-34	92.85	89.82	80.11	70.26	84.19	72.25	75.95	62.60	88.52	81.04	78.03	66.43
	FTransUNet [11]	R50-ViT-B	90.69	86.65	70.91	61.72	83.41	70.28	67.04	56.12	87.05	78.47	68.98	58.92
	MANet [55]	ViT-L	90.51	86.47	70.36	60.78	78.33	60.58	60.30	48.10	84.42	73.53	65.33	54.44
	ASANet [56]	ConvNeXtV2-T	91.39	87.69	76.02	65.49	83.02	69.62	75.52	61.98	87.21	78.66	75.77	63.74
	MRFS [57]	SegFormer-B4	92.35	89.08	77.81	67.70	87.92	78.71	81.96	70.56	90.14	83.90	79.89	69.13
	FDMF-Net (Ours)	SegFormer-B4	**93.48***	**90.69***	**81.16***	**71.57***	87.47*	77.99†	**82.05**†	**70.96**†	90.48	83.34	**81.61**	**71.27**

annealing scheduler. We set the loss-balancing coefficients to $\alpha = 0.10$, $\beta = 0.50$, $\lambda_1 = 0.10$, and $\lambda_2 = 0.01$. In the ASD module, the learnable parameters r_h and r_w for generating frequency-domain masks were initialized to 0.5. All models were trained for 100 epochs with a batch size of 10 and evaluated based on their performance at the 100th epoch. For all comparative approaches, we adopted their officially reported optimal hyperparameter settings. Unless otherwise specified, the same configuration was applied across all subsequent experiments. To demonstrate the robustness of FDMF-Net's performance improvements, we conducted three independent runs with different random seeds for each comparison experiment and reported both the mean and variance ranges. Detailed configurations for each dataset are provided in Table I.

C. Evaluation Metrics

We assess semantic segmentation performance using four widely adopted metrics, i.e., overall accuracy (OA), Kappa coefficient κ (Kappa), mean $F1$ score (mF1), and mIoU. Specifically, OA evaluates both foreground classes and background classes, while mF1 and mIoU are calculated over the foreground classes. Kappa evaluates the agreement between the predicted and ground truth segmentation, accounting for chance agreement. Meanwhile, the segmentation results obtained by different models are visualized to make a qualitative comparison.

To evaluate the computational complexity of FDMF-Net, we examine two complementary metrics: the total number of model parameters and the floating-point operations (FLOPs) count. The former reflects the model's memory footprint, while the latter quantifies its computational workload. An efficient model should minimize both its parameter count and FLOPs.

D. Comparative Experiments

To validate the effectiveness of FDMF-Net, we benchmarked it against 15 segmentation approaches, including seven unimodal methods and eight SOTA multimodal approaches. The inclusion of unimodal methods highlights the added value of multimodal fusion, while the multimodal comparisons serve to evaluate the efficacy of our frequency-domain strategy.

1) Quantitative Accuracy Analysis: Our FDMF-Net showed substantial performance improvements on both WHU-OPT-SAR and YESeg-OPT-SAR, as shown in Table II. FDMF-Net achieved SOTA results across most evaluation metrics, demonstrating effectiveness in complex scenarios. All metrics exhibited low variance, with values below 0.5 across datasets. On WHU-OPT-SAR, the four key metrics exhibited variances below 0.1, indicating both effectiveness and stability. Specifically, FDMF-Net attained an OA of 93.48%, a Kappa of 90.69%, an mF1 of 81.16%, and an mIoU of 71.57%, outperforming existing unimodal and multimodal approaches. Similarly, on YESeg-OPT-SAR, our model achieved 70.16% mIoU and 82.41% mF1, indicating more accurate boundary delineation for small objects and high-precision segmentation under limited data conditions. Although OA and Kappa were slightly lower than some competing methods, this marginal difference was attributed to FDMF-Net's greater sensitivity to minority classes. Rather than inflating performance by favoring dominant categories, our model ensures balanced detection across all semantic regions. Although performance gains over the second-best method on YESeg-OPT-SAR were modest, this was mainly due to its smaller dataset size, which induced greater fluctuations (0.1–0.5). Nevertheless, the results still indicated that our method was capable of delivering stable performance gains. Notably, the mean performance across both datasets exceeded the SOTA. For instance, the Kappa score trailed the strongest baseline CMX by only about 1% while outperforming it in other metrics, demonstrating robust generalization in complex and class-imbalanced settings.

More detailed results on WHU-OPT-SAR are provided in Table III, including per-class accuracies and mean foreground performance. At the category level, FDMF-Net achieved peak

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TABLE III

DETAILED QUANTITATIVE RESULTS (%) ON WHU-OPT-SAR, REPORTING PER-CLASS FOREGROUND SEMANTIC CATEGORY

	Method	Backbone	OA per Category								OA	Kappa	mF1	mIoU
			Far.	Cit.	Vil.	Wat.	For.	Roa.	Oth.	Mean				
MultiModal	MCANet [10]	ResNet-101	88.63	87.86	64.64	91.72	96.11	47.71	7.40	69.15	89.97	85.76	71.11	60.63
	CMFNet [53]	VGG-16	91.32	85.41	77.73	89.06	97.01	48.09	21.26	72.84	91.49	87.85	75.11	64.85
	CMX [54]	SegFormer-B2	93.74	85.55	83.31	91.16	97.00	58.05	29.31	76.87	92.89	89.85	79.18	69.22
	MultiSenseSeg [29]	ResNet-34	91.69	**90.29**	82.96	**92.65**	**97.89**	**68.27**	31.10	79.26	92.85	89.82	80.11	70.26
	FTransUNet [11]	R50-ViT-B	90.87	82.82	76.66	87.26	97.39	49.53	2.24	69.54	90.69	86.65	70.91	61.72
	MANet [55]	ViT-L	89.73	86.52	76.45	88.48	96.65	35.96	5.79	68.51	90.51	86.47	70.36	60.78
	ASANet [56]	ConvNeXtV2-T	92.34	80.65	81.08	89.27	96.60	50.81	21.71	73.21	91.39	87.69	76.02	65.49
	MRFS [57]	SegFormer-B4	93.13	85.13	82.54	90.25	96.83	56.04	25.16	75.58	92.35	89.08	77.81	67.70
	FDMF-Net (Ours)	SegFormer-B4	**94.10***	85.81†	**85.80***	92.33*	97.25*	66.39†	**33.10***	**79.26***	**93.48***	**90.69***	**81.16***	**71.57***

TABLE IV

QUANTITATIVE RESULTS (%) ON ISPRS VAIHINGEN AND ISPRS POTSDAM

	Method	Backbone	Vaihingen				Potsdam				Mean			
			OA	Kappa	mF1	mIoU	OA	Kappa	mF1	mIoU	OA	Kappa	mF1	mIoU
UniModal	SegFormer [48]	SegFormer-B4	90.81	87.96	89.64	82.27	90.72	87.09	91.74	84.75	90.77	87.53	90.69	83.51
	TransUNet [49]	R50-ViT-B	91.59	88.67	90.54	83.11	90.80	87.88	92.01	85.54	91.20	88.28	91.28	84.33
	UNetFormer [50]	ResNet-18	91.11	88.02	89.77	81.86	90.18	87.09	91.43	84.55	90.65	87.56	90.60	83.21
	RS^{3}Mamba [51]	R18-Mamba-T	91.62	88.71	90.36	82.84	90.14	87.03	91.37	84.47	90.88	87.87	90.87	83.66
	PyramidMamba [52]	ResNet-18	87.95	83.77	85.59	75.39	85.00	80.30	86.47	76.59	86.48	82.04	86.03	75.99
	SFFNet [43]	ConvNext-T	90.72	87.51	89.24	80.96	89.09	85.64	90.42	82.89	89.91	86.58	89.83	81.93
	AFENet [20]	ResNet-18	91.74	88.88	90.53	83.07	90.74	87.81	91.93	85.43	91.24	88.35	91.23	84.25
MultiModal	MCANet [10]	ResNet-101	89.69	86.12	87.27	77.96	88.42	84.78	89.27	80.98	89.06	85.45	88.27	79.47
	CMFNet [53]	VGG-16	90.99	87.87	89.52	81.45	89.64	86.37	90.72	83.37	90.32	87.12	90.12	82.41
	CMX [54]	SegFormer-B2	92.16	89.45	90.91	83.72	91.24	88.45	92.36	86.14	91.70	88.95	91.64	84.93
	MultiSenseSeg [29]	ResNet-34	91.65	88.76	89.64	81.71	89.04	85.59	90.45	83.02	90.35	87.18	90.05	82.37
	FTransUNet [11]	R50-ViT-B	92.19	89.47	90.63	83.29	90.61	87.66	91.58	84.79	91.40	88.57	91.11	84.04
	MANet [55]	ViT-L	**92.62**	**90.06**	90.66	83.34	90.42	87.40	91.61	84.84	91.52	88.73	91.14	84.09
	ASANet [56]	ConvNeXtV2-T	90.06	86.62	86.92	77.52	88.89	85.35	89.81	82.03	89.48	85.99	88.37	79.78
	MRFS [57]	SegFormer-B4	91.79	88.95	90.95	83.79	91.11	88.28	92.28	86.02	91.45	88.62	91.62	84.91
	FDMF-Net (Ours)	SegFormer-B4	91.96*	89.17†	**90.97***	**83.82***	**91.52***	**88.83***	**92.55***	**86.45***	**91.74**	**89.00**	**91.76**	**85.14**

accuracies of 94.10% for "Farmland," 85.80% for "Village," and 33.10% for "Others," highlighting its precision and robustness across diverse geographic contexts. Even for fine-grained categories such as "Road," as well as the underrepresented "Others" class, FDMF-Net secured the best or second-best results, averaging about 10% higher than leading multimodal methods, yielding the highest mean OA. Meanwhile, the performance variance remained relatively small for most categories, particularly for the minority classes that our method specifically targets, where the results exhibited greater stability, effectively ruling out the influence of random variations. These findings confirm that our frequency-domain-based feature enhancement and fusion strategy effectively balances broad background regions and fine-grained targets while boosting the discriminability of small-sample classes.

Results on the Vaihingen and Potsdam datasets are summarized in Table IV. On Vaihingen, FDMF-Net improved mIoU and mF1 to 83.82% and 90.97%, respectively, validating its superior performance on small objects and challenging categories in relatively simple scenarios. On Potsdam, which includes larger and more complex scenes, FDMF-Net delivered gains of 86.45% mIoU, 92.55% mF1, 88.83% Kappa, and 91.52% OA. Moreover, FDMF-Net achieved SOTA performance in terms of average metrics across both datasets. These consistent improvements across both datasets confirm the generalizability and adaptability of FDMF-Net to diverse remote sensing scenarios. In addition, our method shows more stable and significant gains on datasets with larger and more complex backgrounds while still delivering satisfactory results on relatively smaller datasets with simpler scenes.

2) Qualitative Accuracy Analysis: We visualized the segmentation results of the compared methods across the four benchmark datasets in Figs. 4 and 5. In WHU-OPT-SAR, fine-scale structures such as "Road" were often missed by existing methods due to their narrow geometry and visual similarity to adjacent "Water" regions. In contrast, FDMF-Net effectively leveraged frequency-domain features to preserve sharp "Road" contours. Moreover, our model refined the boundaries of broad land-cover classes such as "Farmland," maintaining global structure while enhancing local accuracy. In YESeg-OPT-SAR, the boundary between "Roads," "Bare Ground," and "Low Vegetation" was frequently blurred, leading to significant confusion among classes. By jointly exploiting modality-invariant low-frequency cues and modality-specific high-frequency details, FDMF-Net reconstructed faint road segments with greater clarity.

In the Vaihingen dataset, small objects such as "Cars" were easily occluded by "Trees" or shadows, making them challenging to detect. FDMF-Net successfully delineated individual vehicles, even in dense or partially occluded regions, owing to

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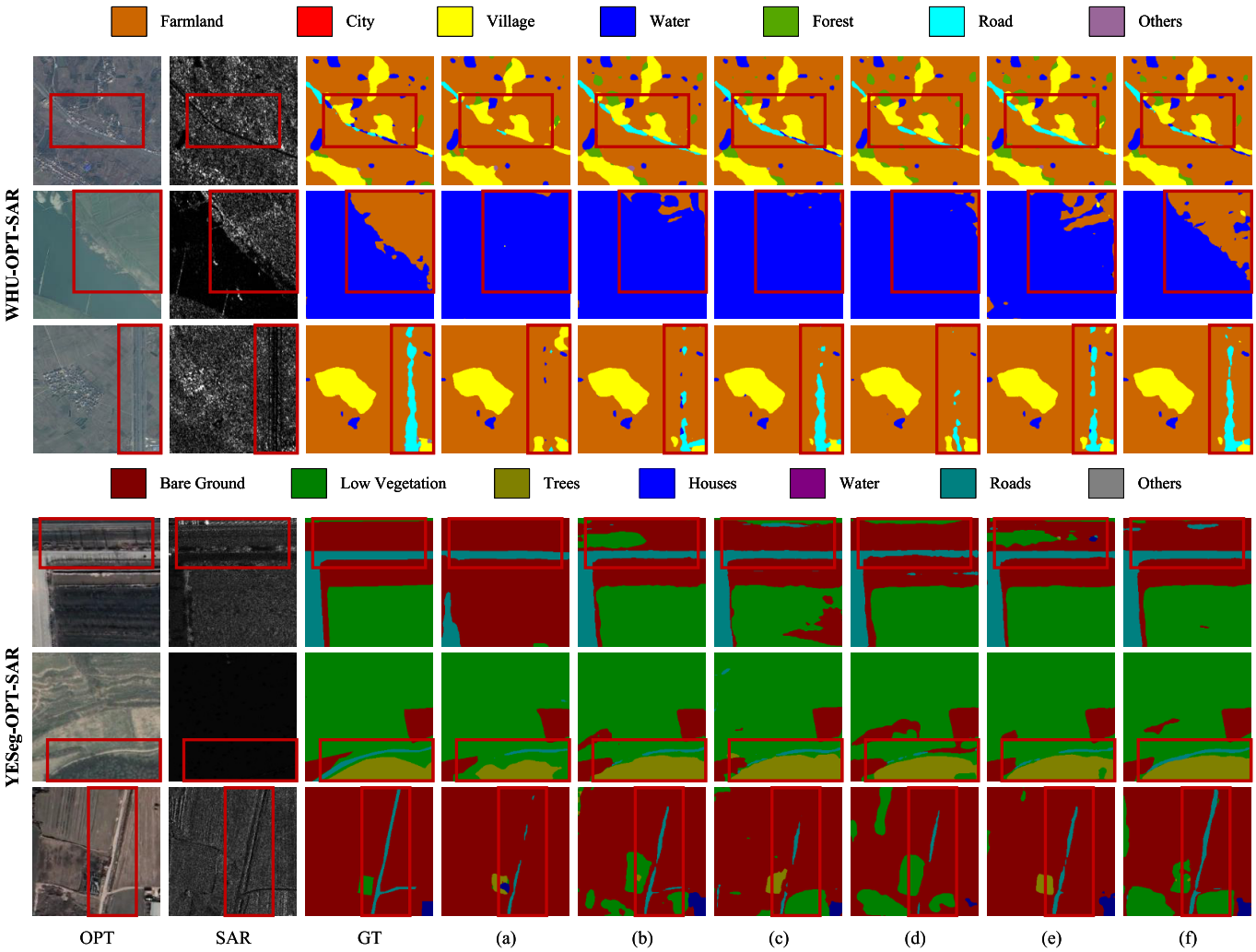


Fig. 4. Qualitative comparison of segmentation results across WHU-OPT-SAR and YESeg-OPT-SAR.Visual results are shown for the leading unimodal baseline (AFENet) and multimodal approaches (CMX, MultiSenseSeg, FTransUNet, and MRFS). (a) AFENet (RGB). (b) CMX. (c) MultiSenseSeg. (d) FTransUNet. (e) MRFS. (f) Our FDMF-Net. Key difference regions are highlighted with red boxes in each subplot.

its deep semantic modeling and enhanced cross-modal fusion. The model also resolved ambiguity between similarly colored classes like "Trees" and "Low Vegetation," producing more accurate segmentation maps. The Potsdam dataset presented even greater challenges due to its complex spatial patterns, with densely interwoven vegetation and a high presence of clutter pixels. Despite these difficulties, FDMF-Net remained robust, accurately segmenting overlapping vegetation types and detecting both large continuous areas and small scattered clutter regions. These results further demonstrate the model's ability to generalize across diverse scenes and object scales, validating the benefits of frequency-aware multimodal feature fusion.

E. Ablation Experiments

We conducted ablation studies on WHU-OPT-SAR to verify the effectiveness of ASD, ME, and LFGF. Furthermore, by testing different hyperparameter combinations, we performed an in-depth analysis of the interactions among the loss components. In addition, we examined the model's behavior under varying input conditions, thereby providing comprehensive evidence of the generalization and transferability of FDMF-Net.

1) Main Modules Analysis: As summarized in Table V, five ablation experiments were designed to evaluate our framework. The baseline model, which includes only the dual-branch encoder and omits the ASD, ME, and LFGF modules, yielded the lowest performance: 69.35% mIoU, 79.17% mF1, 89.90% Kappa, and 92.93% OA. Introducing ASD alone resulted in a modest 0.24% gain in mIoU, indicating that simply decomposing features into frequency components without further refinement brought limited gains. This is primarily due to the significant redundancy between the high- and low-frequency features, which dilutes discriminative capacity. Adding the ME module on top of ASD led to improvements across all metrics. Notably, the accuracy on the scarce "Others" category increased by 1.25%, the "City" class reached 86.22%, and mIoU rose by 0.31%. These gains stem from ME's ability to minimize intramodal mutual information while maximizing intermodal consistency. In contrast, the LFGF module

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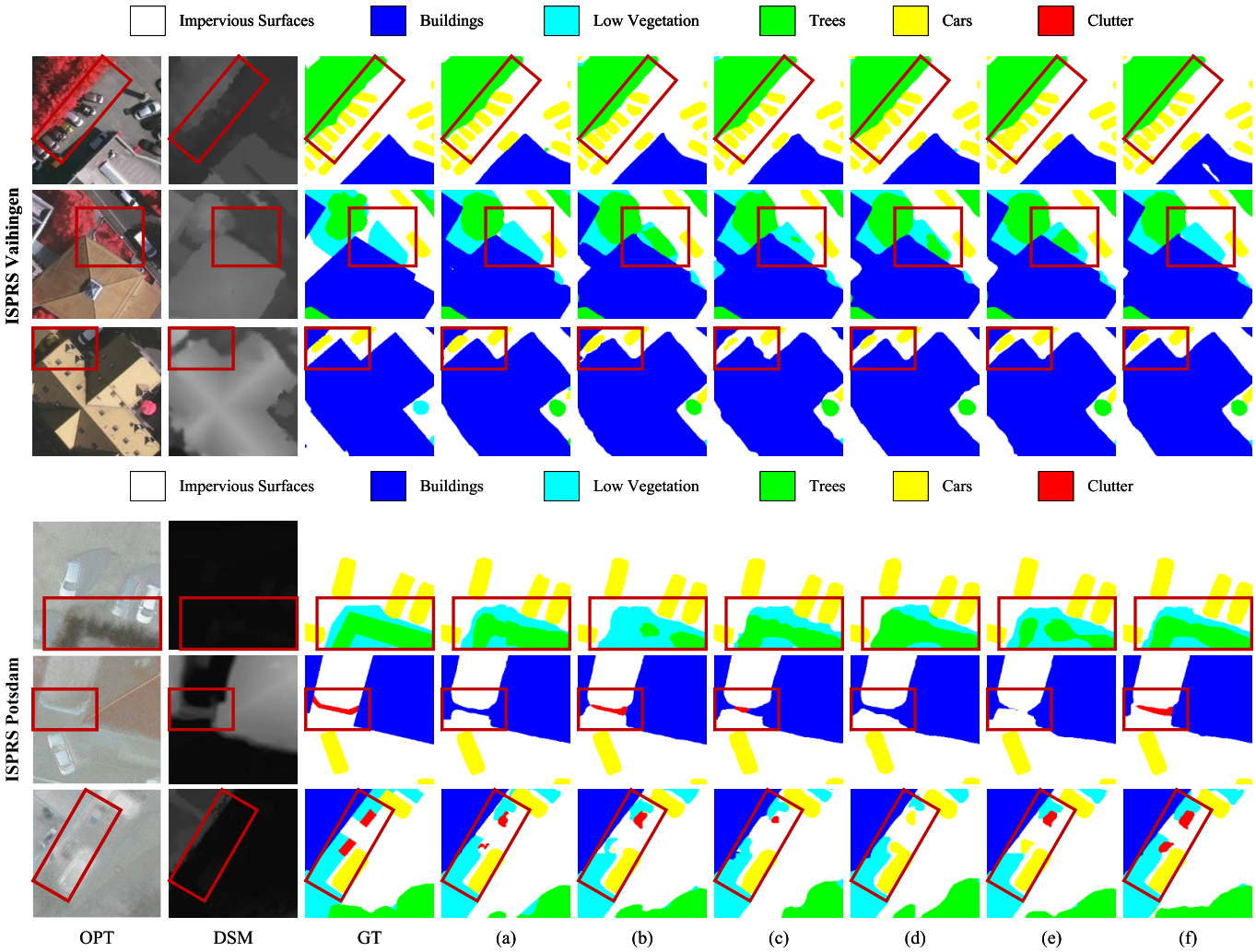


Fig. 5. Qualitative comparison of segmentation results across ISPRS Vaihingen and ISPRS POTSDAM.Visual results are shown for the leading unimodal baseline (AFENet) and multimodal approaches (CMX, MultiSenseSeg, FTransUNet, and MRFS). (a) AFENet (RGB). (b) CMX. (c) MultiSenseSeg. (d) FTransUNet. (e) MRFS. (f) Our FDMF-Net. Key difference regions are highlighted with red boxes in each subplot.

TABLE V
ABLATION RESULTS (%) ON MAIN MODULES ON WHU-OPT-SAR

ID	Modules			OA per Category								OA	Kappa	mF1	mIoU
	ASD	ME	LFGF	Far.	Cit.	Vil.	Wat.	For.	Roa.	Oth.	Mean				
I	✘	✘	✘	93.80	85.92	83.93	91.25	97.08	58.82	26.14	76.71	92.93	89.90	79.17	69.35
II	✔	✘	✘	93.87	86.12	84.22	91.01	97.12	60.17	26.35	76.98	92.98	89.98	79.37	69.59
III	✔	✔	✘	93.63	**86.22**	84.52	91.51	97.13	60.32	27.39	77.25	93.01	90.02	79.47	69.66
IV	✔	✘	✔	**94.31**	85.35	85.35	91.75	97.20	63.46	27.77	77.88	93.31	90.44	80.25	70.64
V	✔	✔	✔	94.09	85.73	**85.75**	**92.54**	**97.19**	**67.95**	**32.63**	**79.41**	**93.47**	**90.68**	**81.24**	**71.66**

employs CA to optimize feature fusion. When combined with ASD, the model achieved 70.64% mIoU, 80.25% mF1, 90.44% Kappa, and 93.31% OA, substantially outperforming the ASD-only variant. This demonstrates that LFGF leverages low-frequency features as modality-invariant cues to guide the fusion of high-frequency modality-specific representations, thereby enhancing both performance and robustness. Integrating ASD, ME, and LFGF into FDMF-Net yielded the best results, 71.66% mIoU, 81.24% mF1, 90.68% Kappa, and 93.47% OA. Compared to the variant without ME, mIoU, mF1, and mean OA each improved by about 1%, and the accuracy on the category of "Others" increased by approximately 5%, surpassing the current SOTA.

The qualitative effect of different module combinations is shown in Fig. 6, alongside a comparison with the next-best multimodal method (MultiSenseSeg). Compared to the baseline model (a), introducing only the ASD module (b) did not improve the extraction accuracy of fine-grained classes such

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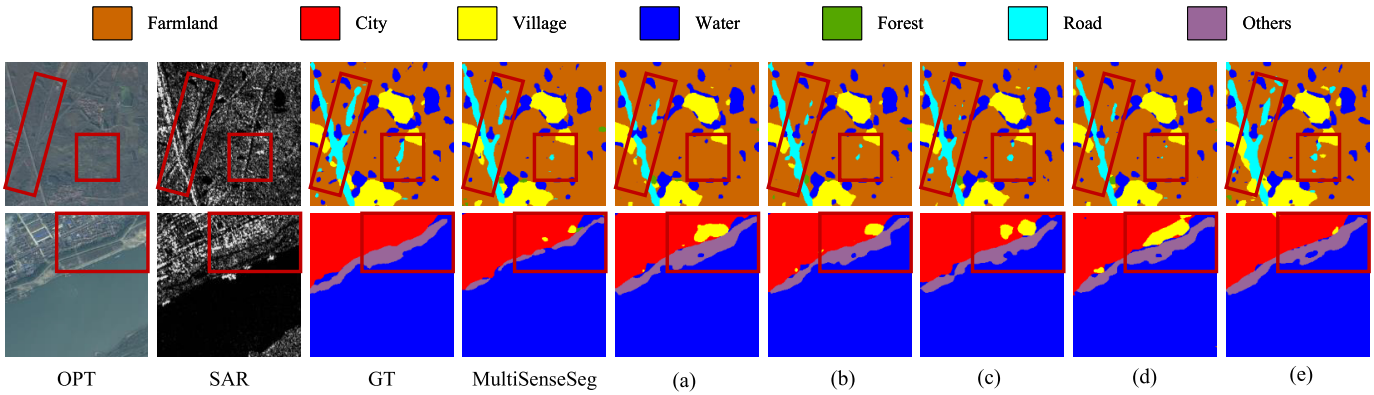


Fig. 6. Visualization of ablation study results on the WHU-OPT-SAR dataset. Comparison between the suboptimal multimodal baseline (MultiSenseSeg) and various ablation variants of FDMF-Net. (a) Baseline, (b) ASD, (c) ASD+ME, (d) ASD+LFGF, and (e) FDMF-Net. Key regions are highlighted with red boxes in each subplot.

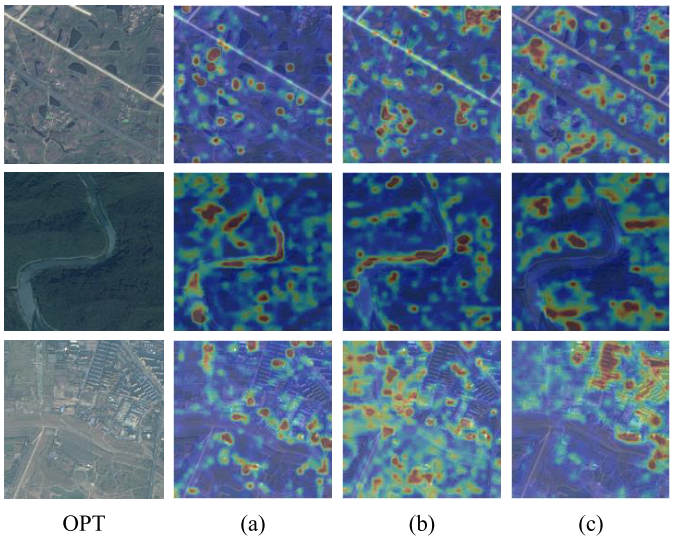


Fig. 7. Qualitative visualization of attention maps in LFGF on three representative scenes from WHU-OPT-SAR. (a) Attention maps from the SA block on low-frequency features. (b) and (c) Attention maps from the CA block that uses low-frequency features as query and high-frequency features from the optical and SAR modalities as key and value, respectively. Warmer colors indicate higher attention weights. Low-frequency guided CA selectively focuses on complementary high-frequency details and exhibits scene-adaptive preferences in both modalities.

as "Road," though it suppressed false positives in confounding regions ("City" and "Village"). Incorporating ASD together with ME (c) or with LFGF (d) yielded a slight enhancement in "Road" segmentation, yet the discriminative power of the remaining individual module still limited overall performance. The full model, i.e., FDMF-Net (e), consistently achieved accurate delineation of both dominant and minority classes.

To rigorously assess the effectiveness of the LFGF module, we conducted a comprehensive ablation on its internal attention designs, as summarized in Table VI. Replacing CA with SA led to a clear drop across all metrics, which indicates that modeling within high-frequency features alone is insufficient for integrating cross-modal details. Among all variants, using the LF, which serves as a shared query to selectively retrieve high-frequency embedding (HF) details via

TABLE VI

ABLATION RESULTS FOR ATTENTION DESIGNS WITHIN THE LFGF MODULE ON WHU-OPT-SAR. WE COMPARE SA AND CA AND TEST DIFFERENT ASSIGNMENTS OF LF AND HIGH-FREQUENCY EMBEDDING (HF) TO THE QUERY (Q) AND THE KEY/VALUE (K/V). THE SYMBOL ‡ MARKS CHANNEL-WISE SHUFFLING OF THE INDICATED FEATURE

Configuration			OA	Kappa	mF1	mIoU
Atten.	Q	K/V				
—	—	—	93.01	90.02	79.47	69.66
SA	HF	HF	93.25	90.27	80.84	70.37
CA	HF	LF	93.19	90.38	80.57	70.23
CA	LF‡	HF	90.99	87.13	74.86	64.26
CA	LF	HF	**93.47**	**90.68**	**81.24**	**71.66**

CA consistently, achieved the best performance. Swapping the roles of query and key/value yielded a consistent degradation, which further suggests that the coarse-to-fine information flow anchored at LF is more stable. Channel-wise shuffling of the LF query caused a pronounced performance collapse, demonstrating that low-frequency guidance is not merely a conceptual description but serves as an effective inductive bias for multimodal fusion. We further presented attention visualizations to provide intuitive evidence, as shown in Fig. 7. Fig. 7(a) shows the SA map within the low-frequency branch, where the model attends to broad structural patterns. Fig. 7(b) and 7(c) shows the CA maps with optical HF and SAR HF features as key/value, respectively. The maps reveal stronger responses on class boundaries and fine textures under the guidance of the LF anchor, such as road and river borders, and within dense building areas. Notably, the two modalities highlight complementary high-frequency cues, consistent with the quantitative results. These observations confirm that the LFGF module not only preserves semantic consistency but also effectively injects discriminative details for fine-grained segmentation.

2) Hyperparameter Analysis: To investigate the impact of hyperparameters α (intramodal mutual information loss weight) and β (intermodal mutual information loss weight) on FDMF-Net's performance, we conducted a sensitivity analysis

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TABLE VII

ABLATION RESULTS (%) FOR THE LOSS-WEIGHT COEFFICIENTS α/β IN THE ME MODULE ON WHU-OPT-SAR

α/β	OA per Category								OA	Kappa	mF1	mIoU
	Far.	Cit.	Vil.	Wat.	For.	Roa.	Oth.	Mean				
0.0 / 0.0	**94.31**	85.35	85.35	91.75	97.20	63.46	27.77	77.88	93.31	90.44	80.25	70.64
0.0 / 0.5	94.01	85.43	84.65	91.49	**97.30**	64.77	31.00	78.38	93.24	90.54	80.91	71.29
0.1 / 0.0	94.01	85.29	85.49	91.60	97.15	64.38	17.98	76.56	93.19	90.27	80.15	70.47
0.5 / 0.5	94.03	85.64	85.74	92.19	97.19	**67.97**	32.21	79.28	93.41	90.60	81.19	71.48
0.2 / 1.0	94.07	85.45	85.75	92.31	97.18	67.14	32.58	79.21	93.46	90.66	81.15	71.57
0.1 / 0.5	94.09	**85.73**	**85.75**	**92.54**	97.19	67.95	**32.63**	**79.41**	**93.47**	**90.68**	**81.24**	**71.66**

TABLE VIII

ABLATION RESULTS (%) FOR THE LOSS-WEIGHT RATIO λ_1/λ_2 BETWEEN THE DICE AND ME LOSSES ON WHU-OPT-SAR

λ_1/λ_2	OA per Category								OA	Kappa	mF1	mIoU
	Far.	Cit.	Vil.	Wat.	For.	Roa.	Oth.	Mean				
0.00 / 0.00	94.16	85.14	85.51	91.93	97.21	63.77	26.99	77.82	93.28	90.41	80.03	70.42
0.00 / 0.01	**94.31**	85.35	85.35	91.75	97.20	63.46	27.77	77.88	93.31	90.44	80.25	70.64
0.10 / 0.00	94.18	85.88	85.65	92.43	97.19	66.97	32.43	79.25	93.47	90.68	81.15	71.55
0.10 / 0.40	94.10	85.75	**86.03**	92.48	**97.23**	67.51	32.55	79.38	**93.50**	**90.72**	81.23	71.63
0.50 / 0.01	93.97	**86.62**	85.53	92.47	97.09	65.98	32.15	79.12	93.41	90.60	80.88	71.26
0.10 / 0.01	94.09	85.73	85.75	**92.54**	97.19	**67.95**	**32.63**	**79.41**	93.47	90.68	**81.24**	**71.66**

TABLE IX

BAND ANALYSIS RESULTS (%) ON WHU-OPT-SAR AND ISPRS POTSDAM, WITH THE SAR MODALITY FIXED WHILE VARYING THE NUMBER OF INPUT OPTICAL BAND SETTINGS

Dataset	Bands	OA per Category								OA	Kappa	mF1	mIoU
		Far.	Cit.	Vil.	Wat.	For.	Roa.	Oth.	Mean				
WHU-OPT-SAR	RGB	93.37	85.23	83.29	90.71	96.97	57.95	24.92	76.06	92.65	89.51	78.29	68.36
	NIRRGB	**94.09**	**85.73**	**85.75**	**92.54**	**97.19**	**67.95**	**32.63**	**79.41**	**93.47**	**90.68**	**81.24**	**71.66**
Dataset	**Bands**	**OA per Category**								**OA**	**Kappa**	**mF1**	**mIoU**
		Imp.	**Bui.**	**Low.**	**Tre.**	**Car.**	**—**	**—**	**Mean**				
ISPRS Potsdam	RGB	93.40	97.98	**88.73**	87.60	**96.72**	—	—	92.89	91.31	88.55	92.40	86.22
	NIRRGB	**93.47**	**98.12**	88.63	**87.78**	96.68	—	—	**92.94**	**91.49**	**88.79**	**92.53**	**86.42**

on the WHU-OPT-SAR dataset, comparing five configurations as shown in Table VII. When $\alpha/\beta = 0.0/0.0$, the ME module is effectively disabled, yielding baseline results of 70.64% mIoU, 80.25% mF1, 90.44% Kappa, and 93.31% OA. Setting $\alpha/\beta = 0.0/0.5$ maximizes intermodal mutual information only; the model still performed well on boundaries and fine details, with the forest category achieving its highest accuracy 97.30%. However, performance on farmland dropped slightly. With $\alpha/\beta = 0.1/0.0$, only intramodal mutual information was minimized, disrupting intermodal consistency and leading to declines across all metrics. The configuration $\alpha/\beta = 0.1/0.5$ delivered the best overall performance by balancing global semantic capture with local detail refinement. In summary, configurations satisfying $\alpha \leq \beta$ outperformed the baseline, with metric fluctuations under 0.3%, demonstrating FDMF-Net's robustness to hyperparameter variations.

We systematically evaluated the effect of the loss weights λ_1 (for the ME loss) and λ_2 (for the Dice loss) on WHU-OPT-SAR, as shown in Table VIII. When training with cross-entropy (0.00/0.00) alone, the model achieved reasonable overall performance but struggled with minority classes. For example, the OA of the category "Others" was only 26.99%. Introducing a moderate ME weight (0.10/0.00) substantially benefited minority categories, raising the OA of "Others" to 32.43%, alongside consistent improvements in global metrics. This indicates that the ME loss effectively facilitates cross-modal disentanglement and complementarity, thereby improving the recognition of challenging and small classes. Adding a small Dice component further stabilized training and gains, achieving the best overall performance while enhancing minority classes. In contrast, using Dice alone without ME yielded only limited gains, highlighting that the performance boost of FDMF-Net primarily stems from the ME loss. When the ME weight was set too high (0.50/0.01), overall performance slightly declined compared with the best configuration, suggesting a tradeoff between segmentation accuracy and representation disentanglement. Overall, FDMF-Net remains robust to variations of λ_1/λ_2 within a reasonable range.

3) Input Band Analysis: We first verified the compatibility of FDMF-Net with different numbers of optical input bands. To ensure compatibility, we adjusted only the input channels of

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TABLE X

ABLATION RESULTS (%) FOR SEGMENTATION PERFORMANCE UNDER VARYING INPUT IMAGE RESOLUTIONS ON ISPRS POTSDAM

Input Size	OA	Kappa	mF1	mIoU
256×256	90.62	87.63	91.28	84.32
512×512	90.89	88.00	91.75	85.16
1024×1024	**90.89**	**88.01**	**91.96**	**85.27**
5000×5000	64.81	53.38	59.11	43.10

the projection layers in the SegFormer backbone, enabling the model to accommodate three-channel (RGB) and four-channel (NIRRGB) inputs without modifying the subsequent encoding and fusion structures. When evaluating the effect of NIR, we kept the SAR (DSM) modality unchanged and compared optical inputs under RGB and NIRRGB configurations, as shown in Table IX.

Compared with RGB, incorporating NIR improved all overall metrics (OA, Kappa, mF1, and mIoU) on both datasets, with particularly significant gains on the more challenging and class-imbalanced WHU-OPT-SAR dataset. At the category level, adding NIR yielded clear benefits for minority and confounding classes, such as "Road" and "Others" in WHU-OPT-SAR. On Potsdam, where class separability is higher and imagery quality is better, the improvements were relatively smaller in magnitude but still consistent. In summary, projection-layer channel alignment enables cross-band compatibility, while using NIR provides additional discriminative cues for specific land-cover classes.

4) Input Size Analysis: We compared performance under different input sizes. Since datasets differ in image size and tiling strategy, we conducted experiments on ISPRS Potsdam (see Table X). Four input sizes were tested: 256×256, 512×512, 1024×1024, and 5000×5000. To accommodate hardware constraints, we fixed the batch size to 5 and reduced the spatial resolution of all projection-layer outputs by half to save memory and GPU resources. For ultralarge 5000×5000 images, bilinear resampling was applied at data loading to resize them to 1024×1024 so that training could be performed within a manageable resource and time budget.

The results indicate that FDMF-Net scales effectively with input resolution. Enlarging the input window provides a richer context, which helps visually similar but semantically distinct regions, while also improving consistency and connectivity. However, performance gains begin to saturate once the input size reaches 1024×1024. This may be attributed to the higher proportion of majority classes within larger windows, which weakens the discriminative signal of minority classes. Under the 5000×5000 setting, performance dropped significantly, mainly because bilinear downsampling erased fine structures and boundary cues. Overall, FDMF-Net benefits from larger input windows, but the substantial resource demands make training on ultralarge images challenging, highlighting an important direction for future work.

F. Mutual Information Estimators Analysis

To assess the adaptability of FDMF-Net to different mutual information estimators, we augmented the ME module with vCLUB-FULL [46] as the intramodal mutual information upper bound and additionally considered the mutual information lower bounds JSD [58] and InfoNCE [59]. We evaluated all pairwise combinations with our current setting, as reported in Table XI. As vCLUB-FULL jointly learns the conditional mean and variance, it is more expressive but less numerically stable, leading to a slight drop in overall performance. Among lower bounds, MINE and InfoNCE were competitive. MINE was the most reliable under our setting, while InfoNCE was more sensitive to hyperparameters such as batch size and temperature. JSD provided the most stable but conservative results due to its looser bound. Performance differences were concentrated on region-level quality (mIoU) and class-balanced accuracy (mF1), suggesting that the estimator choice mainly influences how strongly the model disentangles shared versus complementary cues. Overall, FDMF-Net is robust to the choice of mutual information estimator, with a mild preference for more stable and tighter upper/lower bound pairings that deliver slightly better accuracy.

TABLE XI

ABLATION RESULTS (%) FOR DIFFERENT MUTUAL INFORMATION ESTIMATORS WITHIN THE ME MODULE ON WHU-OPT-SAR, INCLUDING UPPER BOUNDS FOR INTRAMODAL AND LOWER BOUNDS FOR INTERMODAL

Intra-Modal (Upper)	Inter-Modal (Lower)	OA	Kappa	mF1	mIoU
vCLUB-FULL [46]	JSD [58]	93.30	90.43	80.19	70.59
vCLUB-FULL [46]	InfoNCE [59]	93.27	90.40	80.56	70.91
vCLUB-FULL [46]	MINE [47]	93.29	90.41	80.54	70.85
vCLUB-MEAN [46]	JSD [58]	93.30	90.44	80.33	70.72
vCLUB-MEAN [46]	InfoNCE [59]	93.29	90.42	80.47	70.83
vCLUB-MEAN [46]	MINE [47]	**93.47**	**90.68**	**81.24**	**71.66**

TABLE XII

MODEL SCALE ANALYSIS WITH A 256×256 INPUT ON A SINGLE NVIDIA A100 GPU. MIOU IS REPORTED ON WHU-OPT-SAR

	Method	Params (M)	FLOPs (G)	mIoU (%)
UniModal	SegFormer [48]	62.42	11.44	60.00
	TransUNet [49]	105.32	31.14	60.37
	UNetFormer [50]	**11.68**	**5.84**	60.15
	RS^{3}Mamba [51]	43.32	10.04	60.46
	PyramidMamba [52]	125.09	<u>6.92</u>	59.45
	SFFNet [43]	34.18	15.00	60.35
	AFENet [20]	<u>20.24</u>	12.78	59.55
MultiModal	MCANet [10]	70.74	142.38	60.63
	CMFNet [53]	104.07	133.46	64.85
	CMX [54]	66.56	12.42	69.22
	MultiSenseSeg [29]	58.65	128.00	<u>70.26</u>
	FTransUNet [11]	203.56	41.72	61.72
	MANet [55]	317.26	10.62	60.78
	ASANet [56]	82.91	30.30	65.49
	MRFS [57]	134.91	14.76	67.70
	FDMF-Net (Ours)	153.98	20.34	**71.66**

G. Model Complexity Analysis

We evaluated model complexity on a single A100 GPU using input images of size $3 \times 256 \times 256$ and reported mIoU performance on the WHU-OPT-SAR dataset, as displayed in Table XII. The results demonstrate that FDMF-Net

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strikes an excellent balance between computational cost and segmentation accuracy. In terms of parameter count, FTransUNet and MANet have 203.56 and 317.26 M parameters, respectively, while FDMF-Net requires only 153.98 M, indicating a more compact architecture. With only 20.34 G FLOPs, FDMF-Net ranked in the upper mid tier of multimodal methods, demonstrating its relatively low computational overhead. This efficiency is achieved because FDMF-Net rapidly downsamples feature maps in the early stages of the network and places both the Transformer blocks and the spectral–spatial fusion modules at reduced spatial resolutions. As a result, the parameter count stayed moderate while the total FLOPs were kept to a minimum. In comparison, the second-best method MultiSenseSeg relies mainly on convolutional layers and multiscale feature aggregation at higher spatial resolutions, which, despite having fewer parameters, incurs substantially greater computational overhead because of its larger feature maps. Despite this reduced complexity, FDMF-Net achieved the highest mIoU, confirming its computational efficiency. These findings validate that the FDMF-Net design effectively balances performance improvements with model complexity optimization.

V. Conclusion

To accurately capture feature representations in RSIs and integrate them to provide comprehensive semantic information, this study proposes FDMF-Net. This model relies on frequency-domain decomposition and enhances both modality-invariant and specific representations to strengthen the semantic parsing capability of RSIs. We discovered that direct DFT decomposition produces redundant information between frequency features, resulting in a performance bottleneck. Additionally, we designed three modules: ASD, ME, and LFGF, which extract, enhance, and fuse modality representations, respectively. Experiments on four public datasets demonstrate that FDMF-Net outperforms SOTA semantic segmentation methods.

However, the approach still has some limitations. Its relatively large parameter count restricts deployment in resource-constrained environments, and although it reduces computational complexity in certain respects, it remains a potential obstacle for real-time applications. Future work will focus on improving computational efficiency to broaden the model's applicability. Additionally, extending the approach to more diverse and large-scale datasets will validate the generalization capabilities and robustness of the proposed framework.

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